When the Spleen Stops Building: Stromal Remodelling and the Loss of Blood and Immune Cell Production in Spaceflight

The Spleen Machine (OSD-420-3) | Kireeti, Pranav, Aryan, Vedant

Background and rationale. Astronauts return from orbit with weakened immunity. In mice, spaceflight shrinks the spleen and reduces every major splenic leukocyte population [1,2], and the resulting expression profile resembles accelerated immune aging [3]. The spleen organises immune cells into zones and, in mice, supports extramedullary erythropoiesis [4]. Why those leukocytes are lost is unresolved. The cells themselves may be damaged, or the stromal scaffold that organises where blood and immune cells are made may be remodelled. RR-9 spleen was collected through the Biospecimen Sharing Program and has never been analysed in its own right [5].

Preliminary data. Our re-analysis of OSD-420 [6] (RR-9 spleen; 10 flight vs 10 ground-control male C57BL/6J mice, 33 days) identified 80 genes at adjusted $p < 0.05$ in three converging lines. Stromal and perivascular markers rise coordinately (Gpx3, Sparcl1, Pdgfra, Cxcl12). Gpx3 is the strongest signal in the dataset (adjusted $p = 4.8 \times 10\text{-}14$) and has no prior report in any spaceflight tissue. Erythroid and heme-synthesis genes fall as a module (Ank1, Trim10, Gypa, Hmbs), as do lymphoid organisation genes (Rgs13, Ccr7, Madcam1) and type I interferon tone (Isg15, Ifi27l2a). The interferon result replicates Wu et al. [7], who used this dataset to validate human immune-cell findings. Flight and ground cohorts were dissected nine months apart after a hurricane-related rescheduling, treated here as a limitation.

Hypothesis. If spaceflight remodels the splenic stromal niche [8], then flight-exposed spleens will show reduced erythropoietic output and disrupted germinal-centre organisation, and the stromal compartment will change actively rather than shrink in proportion to lost lymphoid mass.

Aim 1. Quantify splenic erythroid progenitors under simulated spaceflight to establish whether reduced erythroid gene expression reflects lost erythropoiesis or altered tissue composition, which bulk RNA-seq cannot distinguish [9,10]. We hypothesise that if spaceflight suppresses splenic erythropoiesis, then perfused spleens from hindlimb-unloaded mice will contain fewer erythroid progenitors per gram of tissue than controls. Progenitors will be staged by Ter119/CD71 flow cytometry, CFU-E colony assays, and histology, with n set by prospective power analysis. Expected outcome: a per-gram progenitor count that separates genuine loss of blood-forming capacity from a composition artifact.

Aim 2. Identify the splenic cell population expressing Gpx3 and test whether its expansion tracks disrupted germinal-centre architecture. Gpx3 rises alongside Pdgfra and Cxcl12 while Ccr7, Madcam1, and Rgs13 fall. We hypothesise that if an expanded perisinusoidal stromal population [8] displaces the gradients required for B- and T-zone segregation, then hindlimb-unloaded mice will form fewer germinal centres after immunisation than controls [11]. Single-nucleus RNA-seq and RNAscope will localise Gpx3, immunofluorescence will map zonal architecture, and NP-KLH immunisation will be scored for germinal-centre number and titre. Expected outcome: the signature assigned to a named cell type, and a direct test of whether remodelling accompanies loss of lymphoid organisation.

Aim 3. Determine whether raising Gpx3 activity preserves splenic erythropoiesis under simulated spaceflight. Gpx3 is a secreted selenium-dependent antioxidant, so dietary selenium offers a direct route to manipulate it. We hypothesise that if elevated Gpx3 marks an oxidative stress response in the splenic niche, then selenium-supplemented hindlimb-unloaded mice will retain more erythroid progenitors than unsupplemented controls. Mice will receive selenium-adequate or supplemented diets through unloading, with Gpx3 activity, oxidative damage, and erythroid output as endpoints. Expected outcome: evidence for whether the stromal signature is protective or pathological, and a dietary countermeasure that can be tested cheaply.

Significance and impact. These aims test where spaceflight immune decline begins, in the immune cells or in the tissue that houses them, and whether a dietary antioxidant can slow it. The same niche processes operate in immunosenescence and chronic-stress immune suppression, so a positive result would extend to aging populations on Earth.

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